Jophin George Joseph, PhD

EXPERIMENTAL & COMPUTATIONAL BIOLOGY | BIOAI | SCIENTIFIC INFERENCE

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github.com/jjbaloo2023 | scholar.google.com/citations?user=1GXfdBAAAAAJ

Experimental and computational biologist connecting cell biology, quantitative imaging, physical models, Bayesian inference, and scientific AI. PhD research combined protein engineering, membrane mechanics, super-resolution microscopy, and image analysis. Extend this approach into industrial bioprocessing and agentic scientific inference.

PROFESSIONAL EXPERIENCE

Senior Scientist | Zoetis Inc.

Kalamazoo, MI | March 2022-Present

- Lead upstream monoclonal-antibody development from clone selection through manufacturing scale. Use AMBR and design of experiments (DoE), scale-down models, bench confirmation, process-characterization support, and technical transfer to investigate scale-dependent biological responses.
- Investigated a CHO metabolic shift at 250 L and larger scales. Reproduced the response in AMBR with an open-pipe sparger after a baffled 5 L model did not. Combined experiments and transport modeling to implicate local hypoxia and support 2,000 L scale-up without the shift.
- Developed a high-throughput plate-based apoptosis assay using Annexin V, propidium iodide, and Caspase-3/7; validated it across three fed-batch media platforms and detected early apoptotic changes before bulk viability declined.
- Built an internally deployed React/FastAPI/Python scale-up application and a GPU-accelerated Python/CuPy transport platform. Link geometry and operating conditions to hydrodynamics, mass transfer, oxygen uptake, shear risk, mixing, and local hypoxia across scales.
- Lead technical coordination and CAD quality review for 100+ bioreactors, mixers, filling lines, and chromatography systems. Use the equipment geometries as a basis for reusable computational experiments and reduced-order models.
- Study a long-acting injectable suspension system spanning formulation and surge vessels, filling equipment, and over 50 m of piping. Use predictions of transport, settling, recirculation, pressure, and dead zones to inform product-uniformity decisions.
- Develop agentic workflows connecting technical records, deterministic analyses, physical models, CFD/HPC tools, and experimental evidence. Retain hypotheses, sources, decisions, outputs, and failures outside chat. Use human scientific review and report when evidence is insufficient.
- Co-lead the computational fluid dynamics (CFD) User Forum. Coordinate cross-functional simulation work and translate mathematical and transport results into experimental decisions.

COMPUTATIONAL RESEARCH

Mechanome | Bayesian and agentic inference for cell mechanics

Open-source research program | 2026-present

Selected participant, Anthropic’s Built with Claude Science Hackathon

- Built an iterative inference framework to compare physical models of membrane and coat mechanics, extending published mechanobiology research. Resolve literature- and AlphaFold/PDB-derived parameters through tool calls, evaluate membrane energetics, retain provenance, and revise models as evidence changes.
- Developed Bayesian inverse inference using dynesty nested sampling and emcee MCMC. Compare model evidence, test identifiability and synthetic recovery, and propose discriminating experiments for underdetermined results. Achieved 96% credible-interval coverage and posterior-median agreement within 0.3%; validated against STED nanotubes (3.8% mean absolute bias) and 2,551 clathrin coats.
- Extend the architecture beyond clathrin-mediated endocytosis to mesoscopic protein-coat interactions and coat dynamics. Compare competing mechanistic representations against published data.

EARLIER RESEARCH EXPERIENCE

Graduate Research Assistant | University of Michigan, Liu Lab

Ann Arbor, MI | September 2016-March 2022

- Engineered epsin domain mutants, truncations, chimeras, fluorescent fusion constructs, and stable cell lines to isolate how structured and intrinsically disordered regions respond to membrane tension during clathrin-mediated endocytosis.
- Built an experimental-computational pipeline combining micropipette aspiration, live-cell SIM-TIRF and other super-resolution imaging, flow cytometry, image segmentation, single-particle tracking, and physical models of membrane mechanics.
- Collaborated with the University of Houston on molecular-dynamics models of tension-sensitive trafficking and with National Taiwan University on membrane tension and macropinocytosis. Designed experiments around model predictions and contributed to peer-reviewed publications.
- Used GenBank, UniProt, PDB, and domain information to design constructs and experiments. Applied qPCR, Western blotting, plate-based assays, and flow cytometry alongside custom Python/MATLAB analysis.

Graduate Intern, Oncology | Amgen Inc.

South San Francisco, CA | January-April 2021

- Developed HTRF and Incucyte live-cell assays to study macrophage interactions with apoptotic tumor cells in the tumor microenvironment.

TECHNICAL SKILLS

Computational biology & inference: Bayesian inference, nested sampling, MCMC, Bayesian evidence, identifiability/degeneracy analysis, mechanistic modeling, synthetic recovery, image segmentation/tracking, collaborative molecular dynamics, PDB/AlphaFold-informed parameterization.

Molecular/cell biology & imaging: molecular cloning, site-directed mutagenesis, stable cell lines, qPCR, Western blot, CHO culture, flow cytometry, HTRF/ELISA; TIRFM, SIM-TIRFM, STORM/PALM, confocal, FRET, live-cell super-resolution imaging, micropipette aspiration.

Software & physical modeling: Python, NumPy/SciPy/pandas/CuPy, MATLAB, R, C++, FastAPI, Docker, Git; CFD (M-Star, ANSYS), membrane energetics, transport phenomena, optimization, regression testing; agentic tool use and evidence-grounded scientific workflows.

EDUCATION & TRAINING

PhD, Mechanical Engineering (Mechanobiology) | University of Michigan | 2022

B.Tech (Hons), Chemical Engineering | IIT Hyderabad | 2016 | Institute Gold Medal; SN Bose Scholar

Fermentation Technology | MIT Professional Education | 2022

SELECTED PUBLICATIONS

Joseph JG et al. Biomechanical role of epsin in influenza A virus entry. Membranes 12(9), 859 (2022).

Joseph JG et al. Complimentary action of structured and unstructured domains of epsin supports clathrin-mediated endocytosis at high tension. Communications Biology 3, 743 (2020).

Joseph JG, Liu AP. Mechanical regulation of endocytosis: new insights and recent advances. Advanced Biosystems 4(5), 1900278 (2020).

Loh J, Chuang MC, Lin SS, Joseph J et al. An acute decrease in plasma membrane tension induces macropinocytosis via PLD2 activation. Journal of Cell Science 132(17), jcs232579 (2019).

Yu ZTF, Joseph JG et al. Centrifugal microfluidics for sorting immune cells from whole blood. Sensors and Actuators B: Chemical 245, 1050-1061 (2017).

Rosselli-Murai LK, Joseph JG et al. The Machado-Joseph disease-associated form of ataxin-3 impacts dynamics of clathrin-coated pits. Cell Biology International 44(5), 1252-1259 (2020).